

RAM NAMBURI

Software Engineer | Backend Systems, Distributed Systems & AI Applications

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PROFESSIONAL SUMMARY

Software engineer with **3 years of professional experience** building backend services, event-driven microservices, cloud-native infrastructure, and AI-driven applications. Experienced with Java, Spring Boot, Kafka, SQL, AWS, Docker, CI/CD, and production operations. Delivers resilient, testable systems from design and implementation through deployment and monitoring.

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript

Frontend: Angular

Backend: Spring Boot, Spring Cloud, Spring Security, FastAPI, REST APIs, JUnit 5, Testcontainers

Distributed Systems: Apache Kafka, microservices, event-driven architecture, API Gateway, idempotency, retries, circuit breakers

Cloud & DevOps: AWS (ECS Fargate, ECR, RDS, Lambda, DynamoDB, S3, CloudWatch, IAM), Docker, Kubernetes, Jenkins, Terraform, CI/CD

Data: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Flyway

AI/ML: LangGraph, LangChain, RAG, LLM integration, ChromaDB, Pinecone, embeddings, scikit-learn, NLP

Tools: Git, Maven, Gradle, Postman, Swagger/OpenAPI, SonarQube, JIRA

PROFESSIONAL EXPERIENCE

Full Stack Software Engineer (Contract) | Optum, UnitedHealth Group — Eden Prairie, MN Jan 2026 – Present

- Built and maintained RESTful Java and Spring Boot microservices for a claims platform processing roughly 2 million daily healthcare transactions; reduced average API response time by 38% through response caching, connection pooling, and query tuning.
- Designed Kafka pipelines that synchronize eligibility and benefits data across six downstream systems, using idempotent consumers and manual acknowledgements to prevent duplicate or lost claim events during retries.
- Implemented OAuth 2.0, JWT authentication, and role-based access control supporting HIPAA and SOC 2 requirements; contributed to an audit with no critical findings.
- Optimized PostgreSQL composite indexes, read-replica routing, and DynamoDB key design to reduce read pressure during peak claim-submission windows.
- Integrated a Python FastAPI NLP service to auto-triage clinical reviews and expanded team adoption of JUnit 5 and Testcontainers integration testing.

Full Stack Software Engineer Co-op | Optum, UnitedHealth Group — Eden Prairie, MN Apr 2025 – Dec 2025

- Contributed to migration of legacy monolithic claims services into containerized microservices deployed on AWS ECS Fargate.
- Helped replace a multi-hour manual deployment workflow with an automated Jenkins CI/CD pipeline measured in minutes.
- Collaborated with backend, platform, and quality-engineering teams to test, document, and release healthcare claims-platform services.

Software Developer | Aryagami Cloud Services — Hyderabad, India Jan 2023 – Dec 2023

- Built and launched a Java 11 and Spring Boot HR and payroll platform comprising four independently deployable microservices—API Gateway, Employee, Salary, and Notification—replacing manual spreadsheet workflows and reducing monthly HR effort by more than 80%.
- Designed Kafka event flows partitioned by employee ID to preserve ordering; added Redis idempotency checks to prevent duplicate payslip delivery under at-least-once processing.
- Strengthened reliability with Resilience4j circuit breakers, retries, and fallback behavior, enabling graceful degradation during downstream-service failures.
- Implemented stateless JWT authentication, Spring Security method-level RBAC, and Flyway versioned migrations for PostgreSQL-backed services.
- Containerized services with multi-stage Docker builds and delivered Jenkins CI/CD pipelines with Testcontainers integration tests, ECR publishing, and rolling AWS ECS Fargate deployments with health-check rollback.

Software Developer Intern | Aryagami Cloud Services — Hyderabad, India Apr 2022 – Jun 2022

- Supported development and testing of Java and Spring Boot backend services for an HR and payroll platform using REST APIs and PostgreSQL.

- Contributed unit tests, API documentation, database tasks, and code-review collaboration within an Agile engineering team.

SELECTED PROJECTS

AI Research Agent (Agentic RAG System) | Python, LangGraph, FastAPI, ChromaDB, OpenAI API, Docker, Railway

- Built and deployed an agentic RAG application that conditionally routes requests between vector retrieval and live web search, grounding responses in cited sources.
- Implemented async token streaming with FastAPI StreamingResponse, semantic chunking and indexing in ChromaDB, source metadata, and dynamic tool selection; delivered sub-200 ms retrieval latency.
- Deployed Dockerized services to Railway and added Redis caching, scheduled Airflow ingestion, and Prometheus monitoring for retrieval quality and token spend.

HR & Payroll Microservices Platform | Java 11, Spring Boot, Kafka, Redis, PostgreSQL, Docker, AWS ECS Fargate

- Designed an event-driven microservices platform with Kafka, Redis idempotency, Resilience4j fault handling, and isolated PostgreSQL data stores.
- Automated integration testing and zero-downtime deployment using Jenkins, Testcontainers, AWS ECR, and ECS Fargate.

Vehicle Service Management System | Python, SQL, C++

- Built a SQL-based backend for concurrent vehicle-maintenance workflows with relational schema design and data-integrity constraints.
- Improved retrieval performance by 15% under simulated load through indexing and query improvements.

Product Review Summarization | Python, NLTK, scikit-learn, AWS

- Built an NLP pipeline for large-scale product-review summarization using distributed AWS processing, reducing processing time by 25%.
- Used a Random Forest model for sentiment classification to generate actionable customer insights.

EDUCATION

University of North Texas — Master of Information Science

Coursework: Advanced Software Engineering, Distributed Systems, Cloud Computing | Thesis: Predictive Software Maintenance Using Machine Learning

GITAM Institute of Technology — B.Tech in Computer Science Engineering

Capstone: Real-Time Traffic Monitoring System Using Java and IoT Sensors

PUBLICATION

Machine Learning Techniques for Software Defect Prediction — International Research Journal of Engineering and Technology (IRJET). Research on predictive analytics and ML-based defect detection to improve software reliability.